

```
!pip install -q transformers torch pillow datasets
```

```
!pip install --no-deps unsloth
!pip install bitsandbytes accelerate peft trl triton cut_cross_entropy
!pip install sentencepiece protobuf datasets huggingface_hub hf_transfer
```

```
Requirement already satisfied: python-dateutil in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.8.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2022.7)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->
Downloading bitsandbytes-0.49.1-py3-none-manylinux_2_24_x86_64.whl (59.1 MB)
59.1/59.1 MB 11.6 MB/s eta 0:00:00
Downloading trl-0.26.2-py3-none-any.whl (518 kB)
518.9/518.9 kB 28.0 MB/s eta 0:00:00
Downloading cut_cross_entropy-25.1.1-py3-none-any.whl (22 kB)
Installing collected packages: cut_cross_entropy, bitsandbytes, trl
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour
unsloth 2026.1.2 requires tyro, which is not installed.
unsloth 2026.1.2 requires unsloth_zoo>=2026.1.2, which is not installed.
unsloth 2026.1.2 requires xformers>=0.0.27.post2; ("linux" in sys_platform or sys_platform == "win32") and (platform_machi
unsloth 2026.1.2 requires datasets!=4.0.*,!=4.1.0,<4.4.0,>=3.4.1, but you have datasets 4.0.0 which is incompatible.
unsloth 2026.1.2 requires trl!=0.19.0,<=0.24.0,>=0.18.2, but you have trl 0.26.2 which is incompatible.
Successfully installed bitsandbytes-0.49.1 cut_cross_entropy-25.1.1 trl-0.26.2
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.1)
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (5.29.5)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: huggingface_hub in /usr/local/lib/python3.12/dist-packages (0.36.0)
Requirement already satisfied: hf_transfer in /usr/local/lib/python3.12/dist-packages (0.1.9)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets) (3.20.2)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.0.2)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.12/dist-packages (from datasets) (4.67.1)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from datasets) (25.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets) (6.0.3)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub) (1.1
Requirement already satisfied: aiohttp!=4.0.0a0,!
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->data
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->data
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->
```

```
!pip install unsloth_zoo
from unsloth import FastVisionModel
import torch
```

```
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets!=4.0.*)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets!=4.0.*)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets!=4.0.*)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->
Downloading unsloth_zoo-2026.1.2-py3-none-any.whl (295 kB)
295.7/295.7 kB 14.0 MB/s eta 0:00:00
Downloading datasets-4.3.0-py3-none-any.whl (506 kB)
506.8/506.8 kB 16.0 MB/s eta 0:00:00
Downloading torchao-0.15.0-cp310-abi3-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (7.2 MB)
7.2/7.2 MB 71.4 MB/s eta 0:00:00
Downloading trl-0.24.0-py3-none-any.whl (423 kB)
423.1/423.1 kB 17.3 MB/s eta 0:00:00
Downloading msgspec-0.20.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (224 kB)
224.9/224.9 kB 12.1 MB/s eta 0:00:00
Downloading tyro-1.0.5-py3-none-any.whl (181 kB)
181.2/181.2 kB 5.9 MB/s eta 0:00:00
Downloading pyarrow-22.0.0-cp312-cp312-manylinux_2_28_x86_64.whl (47.7 MB)
47.7/47.7 MB 18.7 MB/s eta 0:00:00
Installing collected packages: torchao, pyarrow, msgspec, tyro, datasets, trl, unsloth_zoo
Attempting uninstall: torchao
Found existing installation: torchao 0.10.0
Uninstalling torchao-0.10.0:
Successfully uninstalled torchao-0.10.0
Attempting uninstall: pyarrow
Found existing installation: pyarrow 18.1.0
Uninstalling pyarrow-18.1.0:
Successfully uninstalled pyarrow-18.1.0
Attempting uninstall: datasets
Found existing installation: datasets 4.0.0
Uninstalling datasets-4.0.0:
Successfully uninstalled datasets-4.0.0
Attempting uninstall: trl
Found existing installation: trl 0.26.2
Uninstalling trl-0.26.2:
Successfully uninstalled trl-0.26.2
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour
unsloth 2026.1.2 requires xformers>=0.0.27.post2; ("linux" in sys_platform or sys_platform == "win32") and (platform_machi
Successfully installed datasets-4.3.0 msgspec-0.20.0 pyarrow-22.0.0 torchao-0.15.0 trl-0.24.0 tyro-1.0.5 unsloth_zoo-2026.
🐍 Unsloth: Will patch your computer to enable 2x faster free finetuning.
🐍 Unsloth Zoo will now patch everything to make training faster!
```

```
fourbit_models = [
    "unsloth/Llama-3.2-11B-Vision-Instruct-bnb-4bit",
    "unsloth/Qwen2-VL-7B-Instruct-bnb-4bit"
]
```

```
model, tokenizer = FastVisionModel.from_pretrained(
    "unsloth/Qwen2-VL-7B-Instruct-bnb-4bit",
    load_in_4bit=True,
    use_gradient_checkpointing="unsloth",
    device_map="auto",
)
```

```

==(====)== Unsloth 2026.1.2: Fast Qwen2_VL patching. Transformers: 4.57.3.
  \ \ / |
0^0/ \_ / \ Torch: 2.9.0+cu126. CUDA: 7.5. CUDA Toolkit: 12.6. Triton: 3.5.0
 \ \ / |
  \ \ / | Bfloat16 = FALSE. FA [Xformers = None. FA2 = False]
  "-__-" Free license: http://github.com/unslothai/unsloth
Unsloth: Fast downloading is enabled - ignore downloading bars which are red colored!
model.safetensors: 100% 5.90G/5.90G [00:59<00:00, 161MB/s]

generation_config.json: 100% 238/238 [00:00<00:00, 25.3kB/s]

preprocessor_config.json: 100% 567/567 [00:00<00:00, 57.1kB/s]
The image processor of type `Qwen2VLImageProcessor` is now loaded as a fast processor by default, even if the model checkpoint
tokenizer_config.json: 4.30k/? [00:00<00:00, 454kB/s]

vocab.json: 2.78M/? [00:00<00:00, 46.7MB/s]

merges.txt: 1.67M/? [00:00<00:00, 61.5MB/s]

tokenizer.json: 100% 11.4M/11.4M [00:01<00:00, 11.6MB/s]

added_tokens.json: 100% 392/392 [00:00<00:00, 38.6kB/s]

special_tokens_map.json: 100% 614/614 [00:00<00:00, 62.5kB/s]

chat_template.json: 1.05k/? [00:00<00:00, 52.6kB/s]

```

```

model = FastVisionModel.get_peft_model(
    model,
    finetune_vision_layers=True,
    finetune_language_layers=True,
    finetune_attention_modules=True,
    finetune_mlp_modules=True,

    r=16,
    lora_alpha=16,
    lora_dropout=0,
    bias="none",
    random_state = 3407,
    use_rslora=False,
    loftq_config=None
)

```

Unsloth: Making `model.base\_model.model.model.visual` require gradients

```
from torch.utils.data import Dataset
```

```

from torch.utils.data import Dataset
from PIL import Image
import os

class IM2LatexDataset(Dataset):
    def __init__(self, root_dir):
        self.root_dir = root_dir

        # auto-detect txt location
        if os.path.exists(os.path.join(root_dir, "corresponding_png_images.txt")):
            base = root_dir
        elif os.path.exists(os.path.join(root_dir, "data", "corresponding_png_images.txt")):
            base = os.path.join(root_dir, "data")
        else:
            raise FileNotFoundError("Could not find corresponding_png_images.txt")

        self.base = base

        with open(os.path.join(base, "corresponding_png_images.txt")) as f:
            self.image_files = [l.strip() for l in f]

        with open(os.path.join(base, "final_png_formulas.txt")) as f:
            self.formulas = [l.strip() for l in f]

        assert len(self.image_files) == len(self.formulas)

    def __len__(self):
        return len(self.formulas)

```

```
def __getitem__(self, idx):
    image_path = os.path.join(
        self.base,
        "generated_png_images",
        self.image_files[idx]
    )
    image = Image.open(image_path).convert("RGB")

    return {
        "image": image,
        "formula": self.formulas[idx],
    }
```

```
!mkdir -p /content/datasets
```

```
!unzip -q "/content/drive/MyDrive/IMG2LaTeX Zip.zip" -d /content/datasets
```

```
!ls /content/datasets
```

```
PRINTED_TEX_230k
```

```
!ls /content/datasets/PRINTED_TEX_230k
```

```
230k.json          final_png_formulas.txt
corresponding_png_images.txt  generated_png_images
```

```
root_dir = "/content/datasets/PRINTED_TEX_230k"
```

```
dataset = IM2LatexDataset("/content/datasets/PRINTED_TEX_230k")
dataset[0]
```

```
{'image': <PIL.Image.Image image mode=RGB size=200x200>,
 'formula': 'R _ { 1 2 } K _ { 1 } R _ { 2 1 } d K _ { 2 } = d K _ { 2 } R _ { 1 2 } K _ { 1 } R _ { 1 2 } ^ { - 1 } , '}
```

```
instruction = "Write the LaTeX representation for this image."
```

```
def convert_to_conversation(sample):
    return {
        "messages": [
            {
                "role": "user",
                "content": [
                    {"type": "text", "text": instruction},
                    {"type": "image", "image": sample["image"]}
                ],
            },
            {
                "role": "assistant",
                "content": [
                    {"type": "text", "text": sample["formula"]}
                ],
            },
        ]
    }
```

```
sample = dataset[0]
conv = convert_to_conversation(sample)
conv
```

```
{'messages': [{'role': 'user',
 'content': [{'type': 'text',
 'text': 'Write the LaTeX representation for this image.'},
 {'type': 'image', 'image': <PIL.Image.Image image mode=RGB size=200x200>}]},
```

```
{'role': 'assistant',
 'content': [{'type': 'text',
 'text': 'R_{1 2} K_{1} R_{2 1} d K_{2} = d K_{2} R_{1 2} K_{1} R_{1 2} ^{-1}
 ,'}]]}]}
```

```
train_size = 100
test_size = 20

train_dataset = [convert_to_conversation(dataset[i]) for i in range(train_size)]
test_dataset = [
    convert_to_conversation(dataset[i])
    for i in range(train_size, train_size + test_size)
]
```

```
from unsloth import UnslothVisionDataCollator
```

```
data_collator = UnslothVisionDataCollator(
    model = model,
    processor = tokenizer
)
```

Unsloth: Model does not have a default image size - using 512

```
from transformers import TrainingArguments, Trainer
```

```
training_args = TrainingArguments(
    output_dir="./vision2tex",
    per_device_train_batch_size=1,
    gradient_accumulation_steps=8,
    learning_rate=2e-5,
    num_train_epochs=1,
    logging_steps=10,
    save_steps=200,
    fp16=True,
    remove_unused_columns=False,
    report_to="none",
)

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset, # make sure this is correct
    data_collator=data_collator,
)
```

```
trainer.train()
```

```
==(====)= Unsloth - 2x faster free finetuning | Num GPUs used = 1
  \ \ / \  Num examples = 100 | Num Epochs = 1 | Total steps = 13
0^0/ \ \ / \ Batch size per device = 1 | Gradient accumulation steps = 8
 \ \ / \ / Data Parallel GPUs = 1 | Total batch size (1 x 8 x 1) = 8
"-_____" Trainable parameters = 50,855,936 of 8,342,231,552 (0.61% trained)
Unsloth: Will smartly offload gradients to save VRAM!
```

[7/13 00:45 < 00:54, 0.11 it/s, Epoch 0.48/1]

Step Training Loss

[13/13 01:50, Epoch 1/1]

Step Training Loss

10 1.208800

```
TrainOutput(global_step=13, training_loss=1.2148014582120454, metrics={'train_runtime': 202.1368,
'train_samples_per_second': 0.495, 'train_steps_per_second': 0.064, 'total_flos': 592371474819072.0, 'train_loss':
1.2148014582120454, 'epoch': 1.0})
```

```
FastVisionModel.for_inference(model)
```

```
sample = dataset[1]
```

```

messages = [
    {
        "role": "user",
        "content": [
            {"type": "image"},
            {"type": "text", "text": instruction},
        ],
    }
]

inputs = tokenizer.apply_chat_template(
    messages,
    add_generation_prompt=True,
    return_tensors="pt"
)

inputs = tokenizer(
    images=sample["image"],
    text=inputs,
    return_tensors="pt"
).to(model.device)

outputs = model.generate(
    **inputs,
    max_new_tokens=256,
    do_sample=False,
)

decoded = tokenizer.decode(outputs[0], skip_special_tokens=True)

# extract only the assistant's LaTeX part
latex = decoded.split("assistant")[-1].strip()

# remove $$ if present
latex = latex.replace("$$", "").strip()

print(latex)

```

$$E_{\{n\}} - E_{\{m\}} = \left\{ \frac{\lambda (n^2 y^2 - m^2 y^2)}{y^2} \right\}$$

```

def image_to_latex(image):
    FastVisionModel.for_inference(model)

    messages = [
        {
            "role": "user",
            "content": [
                {"type": "image"},
                {"type": "text", "text": instruction},
            ],
        }
    ]

    prompt = tokenizer.apply_chat_template(
        messages,
        add_generation_prompt=True,
        return_tensors="pt"
    )

    inputs = tokenizer(
        images=image,
        text=prompt,
        return_tensors="pt"
    ).to(model.device)

    outputs = model.generate(
        **inputs,
        max_new_tokens=256,
        do_sample=False,
    )

    return tokenizer.decode(outputs[0], skip_special_tokens=True)

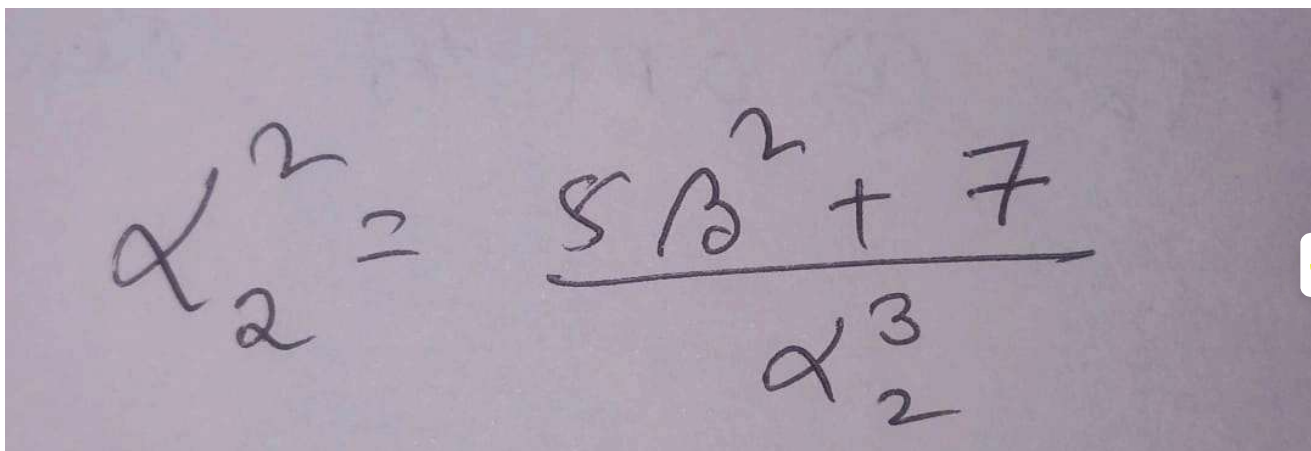
```

```
!find /content/datasets -name "000a20147419b28.png"
```

```
/content/datasets/PRINTED_TEX_230k/generated_png_images/000a20147419b28.png
```

```
from PIL import Image
from IPython.display import display

#image_path = "/content/datasets/PRINTED_TEX_230k/generated_png_images/000a20147419b28.png"
image_path = "/test 1.jpeg"
image = Image.open(image_path).convert("RGB")
display(image)
```



```
latex_output = image_to_latex(image)
latex_output = latex_output.split("assistant")[-1]
latex_output = latex_output.replace("$", "").strip()
print(latex_output)
```

```
x _ { 2 } ^ { 2 } = \frac { 5 \backslash beta ^ { 2 } + 7 } { x _ { 2 } ^ { 3 } }
```

```
model.save_pretrained("model_save")
tokenizer.save_pretrained("model_save")
```

```
[]
```

```
model, tokenizer = FastVisionModel.from_pretrained(
    "model_save",
    load_in_4bit=True,
    use_gradient_checkpointing="unsloth",
    device_map="auto",
)

model = FastVisionModel.get_peft_model(
    model,
    finetune_vision_layers=True,
    finetune_language_layers=True,
    finetune_attention_modules=True,
    finetune_mlp_modules=True,

    r=16,
    lora_alpha=16,
    lora_dropout=0,
    bias="none",
    random_state = 3407,
    use_rslora=False,
    loftq_config=None
)
```

```
import re

def normalize_latex(tex):
    tex = tex.lower()
    tex = tex.replace("$", "")
```

```

tex = tex.replace("#", "/")
tex = tex.replace("\\", "")
tex = tex.replace("\\!", "")
tex = tex.replace("\\left", "").replace("\\right", "")
tex = re.sub(r"\s+", "", tex) # remove spaces
tex = re.sub(r"{}", "", tex) # remove braces
return tex

```

```

preds = []
refs = []

for i in range(len(test_dataset)):
    sample = test_dataset[i]

    image = sample["messages"][0]["content"][1]["image"]
    gt = sample["messages"][1]["content"][0]["text"]

    pred = image_to_latex(image)

    preds.append(normalize_latex(pred))
    refs.append(normalize_latex(gt))

```

```
!pip install -q sacrebleu
```

```
!pip install -q python-Levenshtein
```

```

# 1. Exact Match Accuracy (EMA)
def exact_match_accuracy(preds, refs):
    return sum(p == r for p, r in zip(preds, refs)) / len(refs)

# 2. Expression Recognition Rate (ERR)
def normalize_latex(tex):
    tex = tex.replace(" ", "")
    tex = tex.replace("{", "").replace("}", "")
    tex = tex.replace("\\left", "").replace("\\right", "")
    return tex

def expression_recognition_rate(preds, refs):
    return sum(
        normalize_latex(p) == normalize_latex(r)
        for p, r in zip(preds, refs)
    ) / len(refs)

# 3. BLEU score
import sacrebleu

def bleu_score(preds, refs):
    refs = [[r] for r in refs]
    return sacrebleu.corpus_bleu(preds, refs).score

# 4. Normalized Edit Distance (NED)
import Levenshtein
import numpy as np

def normalized_edit_distance(preds, refs):
    return np.mean([
        Levenshtein.distance(p, r) / max(len(r), 1)
        for p, r in zip(preds, refs)
    ])

```

```

ema = exact_match_accuracy(preds, refs)
err = expression_recognition_rate(preds, refs)
bleu = bleu_score(preds, refs)
ned = normalized_edit_distance(preds, refs)

print(f"Exact Match Accuracy (EMA): {ema:.4f}")

```

```
print(f"Expression Recognition Rate (ERR): {err:.4f}")  
print(f"BLEU Score: {bleu:.2f}")  
print(f"Normalized Edit Distance (NED): {ned:.4f}")
```

Mr F

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